

1. Import Libraries

- ``pandas``: For handling structured data like CSV files
- ``numpy``: For numerical operations
- ``matplotlib.pyplot``, ``seaborn``: For plotting and visualizations
- ``LinearRegression``: To create and train a regression model
- ``train_test_split``: To split the dataset into training and testing sets
- ``mean_squared_error``, ``r2_score``: To evaluate model performance

2. Kaggle API + Dataset Load

- ``files.upload()``: Uploads your Kaggle API key file (kaggle.json)
- Shell commands download the dataset using the Kaggle API
- ``read_csv``: Loads the dataset into a DataFrame for analysis

3. Explore the Dataset

- ``info()``: Displays data types and non-null counts
- ``describe()``: Summary statistics for numerical columns
- ``isnull().sum()``: Checks for missing values
- ``select_dtypes(np.number)``: Filters only numeric columns
- ``corr()``: Computes pairwise correlation of columns
- ``sns.heatmap(...)``: Visualizes correlation matrix to find relationships

4. Preprocess the Data

- ``drop('Car_Name')``: Removes non-informative column
- ``pd.get_dummies(..., drop_first=True)``: Converts categorical columns to numerical dummy variables while avoiding multicollinearity

5. Split Features and Target

- ``X = df.drop('Selling_Price')``: Inputs (features)
- ``y = df['Selling_Price']``: Output (target variable)

6. Train-Test Split

- `train_test_split(...)`: Divides data into training and test sets (80/20)
- `random_state=42`: Ensures consistent results across runs

7. Train the Model

- `LinearRegression()`: Initializes the model
- `.fit(X_train, y_train)`: Trains the model using the training data

8. Predict and Evaluate

- `.predict(X_test)`: Makes predictions using the trained model
- `mean_squared_error`: Measures average squared error between predicted and actual values
- `r2_score`: Measures how well the model explains variation in the target variable (closer to 1 is better)